NATIONAL MATHEMATICS CONTEST

O-LEVEL PAPER 1.

SATURDAY 6TH APRIL, 2019 9:00 AM - 12:15 PM

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

(a) This competition is conducted on the assumption that proper security is maintained. UMS reserves the right, should there be evidence or suspicion of any malpractice, to reject scripts from a particular school/institution.

(b) There are **two** *sections and each section carries* 50 *marks. Attempt all questions in exactly three hours fifteen minutes (*$9:00am$ *to* $12:15pm$*). Marks will be awarded for only answers for which a clear and logical layout of the working is provided.*

(c) Indicate your names, gender, class, school and district ***on all*** *your answer sheets.*

(d) Answer scripts of only registered students should be returned to the contest coordinator, latest **Monday 8th April, 2019**.

(e) **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 27*th July,* 2019 *at NOON at Makerere University, Kampala.*

(f) National Mathematics Contest 2019 *Paper* 2 *will be done on* 29*th June,* 2019 *at various centres.*

SECTION A : 5 marks each

1. If we mix one litre of lemonade that contains 4% lemon with two litres of lemonade that contains 10% lemon, what is the percentage of lemon in the resulting three litre mixture?

2. If $\frac{1}{x} - \frac{1}{x+1} = \frac{1}{36290}$, what is the value of the positive integer x?

3. Peter's age is the sum of the ages of his two brothers. Six years ago, his age was the product of the ages of his two brothers. How old is Peter and his brothers?

4. Seera's score on a mathematics test was recorded incorrectly by the teacher. Her real score was exactly four times the score that the teacher recorded. When the teacher corrected her mistake, the average score of the class went up by 2. There are 24 students in Seera's class (including Seera). What was Seera's real score on the mathematics test?

5. Suppose you begin counting the numbers $1, 2, 3, 4, 5$ starting with 1, then $2, 3, 4, 5$ and back to 1 as in $1, 2, 3, 4, 5, 4, 3, 2, 1, 2, 3, 4, 5, 4, 3, 2, 1, \ldots$. What is the 2019th number you count?

6. John and Brenda play a game where they toss a coin that is equally likely to come up heads or tails. They take turns flipping the coin, with John going first. The first person to flip tails wins assuming that the result of John's toss is revealed after Brenda's toss, determine the probability that John wins the game.

7. An employer reduced the salary of his employees by 20%. The employees complained. After a month he increased the new salary by 20%. One of the employees still complained. What was the reason for the complaint?

8. What is the value of x in $5^x - 5^{x-2} = 120\sqrt{5}$?

9. The triangle ABC has edge lengths $BC = 20$, $CA = 21$, and $AB = 13$. What is its height h shown in the Figure 1?

10. Alice is driving to Bob's house, intending to arrive at a certain time. If she drives at 60 km/h she will arrive 5 minutes late. If she drives at 90 km/h she will arrive 5 minutes early. Determine the speed at which she should drive to arrive exactly on time.

SECTION B : 10 marks each

11. Okello and Mugisha decide to race each other on an 800 metre track. Since Mugisha is faster than Okello, the two decided to give Okello a head start. In the first race, Okello was given a 20 metre head start, however, Mugisha still won and finished 2 seconds earlier than OKello. In the second race, Okello was given a 38 metre head start, and this time Okello won and finished 1 second ahead of Mugisha. Assuming both Okello and Mugisha ran at uniform speeds in both races, determine the speeds (in metres per second) of both runners.

12. In a sequence, each term after the first is the sum of squares of the digits of the previous term. For example, if the first term is 42 then the next term is $4^2 + 2^2 = 20$. The next term after 20 is then $2^2 + 0^2 = 4$, followed by $4^2 = 16$, which is then followed by $1^2 + 6^2 = 37$, and so on, giving the sequence $42, 20, 4, 16, 37$, and so on.

    (a) If the first term is 44, what is the 2019th term?

    (b) If the first term is 25, what is the 2019th term?

13. The number 102564 has the property that if the last digit is moved to the front, the resulting number, namely 410256, is 4 times larger than the original number: $410256 = 4 \times 102564$. Find a six-digit number whose last digit is 9 and becomes 4 times larger when we move this 9 to the front.

14. A 3×3 magic square consists of nine different numbers placed in a grid in such a way that the sum of the numbers in each row, each column, and the two main diagonals is the same. The numbers $5, 10, 15, 20, 25, 30, 35, 40$, and 45 are used to form a magic square. Some of these numbers have been placed, as shown. Find the possible value(s) of X and Y.

20	45	a
		X
		Y

15. The Figure 2 below shows an 8 by 9 rectangle cut into three pieces by two parallel slanted lines AE and BD. The three pieces all have the same area. How far apart are the slanted lines?

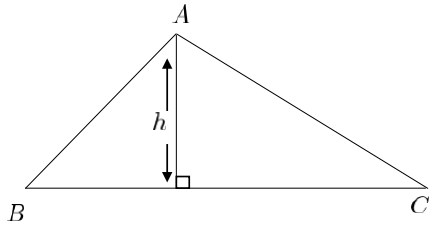


Figure 1: Qn 9 figure

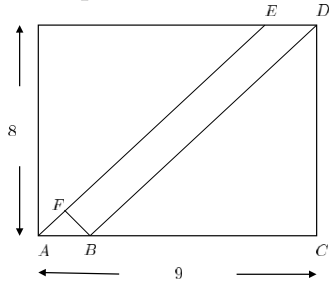


Figure 2: Qn 15 figure

NATIONAL MATHEMATICS CONTEST
O-LEVEL PAPER 1 SOLUTIONS.

SATURDAY 6TH APRIL, 2019 9:00 AM - 12:15 PM

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SECTION A : 5 marks each

1. Let the quantity of Lemon in 1 litre be 4 units. Then the quantity of lemon units in 2 litres is 20. So the total number of Lemon units in the 3 litres is 24. Hence the percentage of lemon in 3 litre mixture is $\frac{24}{3}\% = 8\%$.

2. $\dfrac{1}{x} - \dfrac{1}{x+1} = \dfrac{1}{36290} \quad \Rightarrow \quad x(x+1) = 36290$ therefore $x = 190$.

3. Let a and b be the ages of Peter's two brothers six years ago. Where we assume that $a \geq b$. Then Peter's age six years ago was ab, so his age now is $ab + 6$. But his brothers' ages now must be $a + 6$ and $b + 6$ so Peter's age now is $a + 6 + b + 6 = a + b + 12$. This implies that $ab+6 = a+b+12 \Rightarrow ab-a-b = 6$. If we add both sides by 1 and factor the resulting equation gives, $(a-1)(b-1) = 7$. The only integer solution to this equation that satisfy $a \geq b > 0$ is when $a - 1 = 7$ and $b - 1 = 1$ thus $a = 8$ and $b = 2$. These were the ages of the brothers six years ago so the brothers are now 14 and 8 respectively $\Rightarrow$ Peter's age is 22 years.

4. Since the average score of the class of 24 students went up by 2 when the teacher corrected Seera's mistake, the total score of all students must have gone up by $2 \times 24 = 48$ points. Thus 48 points must represent the difference between the originally recorded score and four times that score that is 48 must equal to exactly three times the originally recorded score. So the originally recorded score must have been $\dfrac{48}{3} = 16$. So Seera's real score must have been $16 \times 4 = 64$.

5. The following pattern appears while counting:

$$1,\ 2,\ 3,\ 4,\ 5,\ 4,\ 3,\ 2,\ 1,\ 2,\ 3,\ 4,\ 5,\ 4,\ 3,\ 2,\ 1,\ 2,\ 3,\ 4,\ 5,\ 4,\ 3,\ 2,\ 1,\ldots$$

We notice that 1 appears at the first, ninth, 17th, 25th, ... position, so 1 also appears at the 2017th position $\Rightarrow$ there is a 2 at 2018th and 3 appears at 2019th.

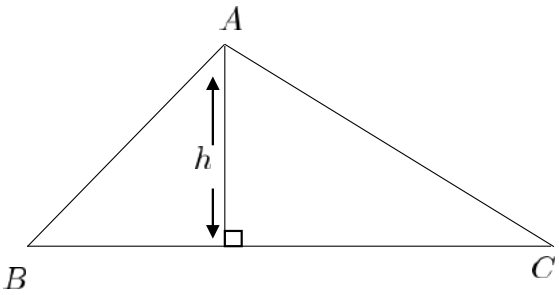
6. Let T denote tails and H denote heads. Suppose John rolls/tosses the coin but the result of his roll is revealed only after Brenda has rolled. The four equally likely outcomes in the sample space are TT, TH, HT and HH. If the fourth outcome occurs, then the game is a draw, so they play again. For John to win the first two events have to occur out of the three possible events thus $\frac{2}{3}$ is the required probability.

7. Let x be the original salary, decreased by $20\% \Rightarrow 0.8x$ of the salary is paid and this is increased by 20% to $0.8 \times 1.2 \times x = 0.96x \Rightarrow 96\%$ of the original salary is got meaning it is still less than the original so the employees should continue to complain.

8.

$$5^x - 5^{x-2} = 120\sqrt{5} \quad \Leftrightarrow \quad 5^x\left(1 - \frac{1}{25}\right) = 24 \times 5^{\frac{3}{2}} \quad \Leftrightarrow \quad 5^x\left(\frac{24}{25}\right) = 24 \times 5^{\frac{3}{2}} \quad \Leftrightarrow \quad x = \frac{7}{2}.$$

9. From the triangle ABD



we have

$$x^2 + h^2 \;=\; 169 \tag{1.1}$$

and from triangle ACD we have

$$21^2 \;=\; (20 - x)^2 + h^2 \tag{1.2}$$

Solving (1.1) and (1.2) gives $x = 3.2$. Thus from (1.1) we have $h = \sqrt{13^2 - x^2} = 12.6$

10. Let D and V be distance and velocity, respectively. Let T denote the exact time and let x denote the velocity corresponding to that time. Noting that 5 minutes is $\frac{1}{12}$ of an hour. For the late journey we have

$$\frac{D}{60} \;=\; T + \frac{1}{12}, \tag{1.3}$$

For the early Journey we have

$$\frac{D}{90} \;=\; T - \frac{1}{12} \tag{1.4}$$

and for the exact journey we have

$$\frac{D}{x} = T \Rightarrow D = Tx \tag{1.5}$$

From (1.3) and (1.4) we get $x = 72km/hr$.

SECTION B : 10 marks each

11. Suppose Mugisha ran 800 m in t seconds. Then Okello ran 780 m in $t+2$ seconds and 762 m in $t-1$ seconds. Since Okello ran at uniform speed in both races we have

$$\frac{780}{t+2} = \frac{762}{t-1} \Rightarrow t = 128sec.$$

Therefore Okello runs at $\frac{780}{130} = 6m/s$ and Mugisha runs at $\frac{800}{128} = 6.25m/s$.

12. (a) If the first term is 44, what is the 2019th term? The following are the first few terms of the sequence:

$$\begin{aligned} & 44, \\ 4^2+4^2 &= 32, \\ 3^2+2^2 &= 13, \\ 1^2+3^2 &= 10, \\ 1^2+0^2 &= 1, \\ 1^2 &= 1, \\ 1^2 &= 1, \\ & \vdots \end{aligned}$$

This implies that the sequence stablises at 1 and the 2019th term is 1.

(b) If the first term is 25, what is the 2019th term? The following are the first few terms of the sequence:

$$\begin{aligned} & 25, \\ 2^2+5^2 &= 29, \\ 2^2+9^2 &= 85, \\ 8^2+5^2 &= 89, \\ 8^2+9^2 &= 64+81 = 145, \\ & \vdots \end{aligned}$$

This sequence simplifies to

$$25,\ 29,\ 85,\ 145,\ 42,\ 20,\ 4,\ 16,\ 37,\ 58,\ 89,\ 145,\ 42,\ 20,\ 4,\ 16,\ 37,\ 58,\ 89, \ldots$$

So the 8th, 16th, 24th, ..., 2016th term is 4 thus the 2017th term is 16, 2018th is 37 and the 2019th term is 58.

13. We need to find a number a, b, c, d, e so that,

$$\begin{array}{cccccc} a & b & c & d & e & 9 \\ & & & & \times & 4 \\ \hline 9 & a & b & c & d & e \\ \hline \end{array}$$

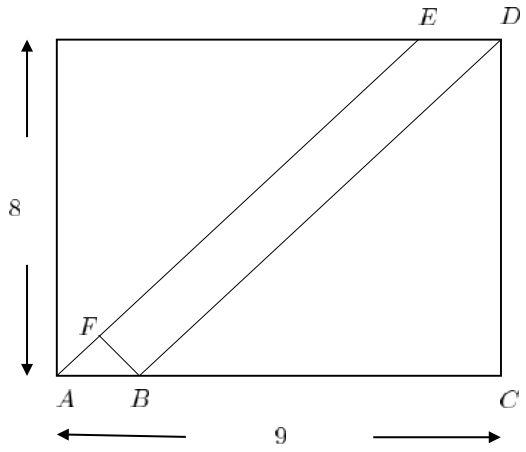
This implies that $e = 6, d = 7, c = 0, b = 3$ and $a = 2$ thus the six digit number whose last digit is 9 and becomes 4 times larger when we move 9 to the front is 230769.

14. For

20	45	a
		X
		Y

The sum of all numbers is $5 + 10 + 15 + 20 + \ldots + 45 = 225$. The sum of three rows and three columns is the same that is sum of three rows and/or three columns is $\dfrac{225}{3} = 75$. The number in the right upper corner is $75 - (20 + 45) = 10 = a$. Thus $75 - (X + Y) = 10$, possible values of X and Y are $(35, 30)$ or $(30, 35)$.

15. Given the figure,



Since the area of the rectangle is $8 \times 9 = 72$, the area of each of the three pieces must be $\dfrac{72}{3} = 24$. In particular the area of right triangle BCD must be 24. Since $CD = 8$ and the area of $\triangle BCD$ is

$$\frac{1}{2}BC \times CD = BC \times 4 \Rightarrow BC = \frac{24}{4} = 6.$$

By Pythagoras Theorem $BD = \sqrt{6^2 + 8^2} = 10$. Since the area of the parallelogram $ABDE = 24$ and is given by $24 = BD \times BF$ then $BF = \dfrac{24}{10} = 2.4$.